

Ryan Mahony Holmes, Ph.D.

Bureau of Meteorology – Sydney – AUSTRALIA

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Education

Doctor of Philosophy, GPA: 4.08/4 <i>Stanford University</i>	2011–2016 <i>Stanford, CA, United States</i>
Bachelor of Philosophy (Physics), First class honours 95/100 with University medal <i>Australian National University (ANU)</i>	2006–2010 <i>Canberra, Australia</i>

Employment

Climate Scientist <i>Bureau of Meteorology</i> Research Scientist (EL1) in the Seasonal and Marine Applications Team, Science and Innovation Group. Applied research developing seasonal forecast products in the marine and coastal sectors. Developed a probabilistic blended seasonal forecast system for coastal sea level for the Australian Climate Service. Additional contributions to uplifting the Bureau's storm surge forecast system, seasonal forecast verification and independent research.	April 2023–present <i>Sydney, Australia</i>
ARC Discovery Early Career Researcher Award (DECRA) Fellow <i>University of Sydney, School of Geosciences</i> Academic level B research fellow position funded by ARC DECRA award DE21010004, “Mixing and air-sea coupling in the Pacific: Toward better El Niño forecasts”. Independent research in regional coupled climate modelling to understand the impact of small-scale ocean mixing and mesoscale air-sea coupling processes on Pacific climate variability and predictability.	August 2021–March 2023 <i>Sydney, Australia</i>
Senior Research Associate <i>University of New South Wales (UNSW)</i> Two 0.5-FTE academic level B postdoc positions in the School of Mathematics and Statistics and the ARC Centre of Excellence for Climate Extremes. Led international collaborative projects on ocean heat transport, climate variability and ocean modelling. Co-supervision of Ph.D. and masters students.	2018–2021 <i>Sydney, Australia</i>
Research Associate <i>University of New South Wales</i> Academic level A postdoc position in the Climate Change Research Centre working with Professor Matthew England on ocean heat transport, the role of stochastic oceanic variability in El Niño and Antarctic coastal circulation.	2016–2017 <i>Sydney, Australia</i>
Postdoctoral Fellow <i>University of New South Wales</i> Six-month academic level A postdoctoral position in the School of Mathematics and Statistics working with Professor Trevor McDougall. Contributed to a high-impact study on abyssal ocean circulation (de Lavergne et al. 2017, <i>Nature</i>).	2016 <i>Sydney, Australia</i>
Graduate Researcher <i>Stanford University</i> Graduate thesis project on equatorial ocean dynamics and regional ocean modelling. 5 (4 first-author) journal publications.	2011–2016 <i>Stanford, CA, United States</i>

Key Skills & Strengths

- Broad knowledge of interdisciplinary oceanography and geophysical fluid dynamics
- Extensive observational, theoretical and numerical-modelling based ocean research experience
- Ability to work collaboratively and independently in both academic and applied research settings
- Excellent written and oral communication skills with an outstanding publication record for career stage

Technical skills and experience:

- Development and verification of seasonal forecast products for the marine and coastal sectors.
- Code development and scientific contributions to numerical ocean and climate models.
- Data analysis of large climate model and observational datasets.

- o **Languages and software:** Fortran, Python, MatLab, Git, GitLab, UNIX, BASH, PBS.
- o **High Performance Computing:** National Computational Institute (NCI) Gadi and associated systems.

Research Outputs and Impact

Academic Record and Selected Publications

I have published 53 peer-reviewed journal articles (+4 in review), including 19 as first-author and 10 as second-author, which have received 1608 citations with a h-index of 24 (Google Scholar). Typically, top oceanographers and climate scientists in Australia have h-indexes equal to or higher than their years post Ph.D.. **On this metric I am exceptional with an h-index (24) more than twice my years post Ph.D. (10).** All except 1 of my papers are published in journals with Q1 quartile rankings (top 25% of journal in relevant subject category), including high-impact publications in journals such as *Nature* (3), *Nature Climate Change* (1) and *Geophysical Research Letters* (4). 3 of my papers were in the top 1% of cited papers in their field for their year (Web of Science). **Selected list of key publications** (see <https://rmholmes.github.io/publications/> for a complete list)

- o **Holmes, R.**, Smith, G. and Spillman, C. (2025) Seasonal sea level forecasts for the Australian coast. *Journal of Southern Hemisphere Earth System Science*, 75, ES24047.
- o **Holmes, R.**, Richet, O., Hague, B., Smith, G. and Spillman, C. (2025) Seasonal outlooks for coastal high tide flooding. *Natural Hazards*
- o England, M., Li, Z., Huguenin, M., Kiss, A., Sen Gupta, A., **Holmes, R.** and Rahmstorf, S. (2025) Drivers of the extreme North Atlantic marine heatwave during 2023. *Nature*, 642, 636-643.
- o **Holmes, R.**, Zika, J., Griffies, S., Hogg, A., Kiss, A. and England, M. (2021) The geography of numerical mixing in a suite of global ocean models. *Journal of Advances in Modeling Earth Systems*, e2020MS002333.
- o **Holmes R.**, Zika, J. D. and England M. H. (2019) Diathermal Heat Transport in a Global Ocean Model, *J. Phys. Oceanogr.*, **49**, 141-161. **[top 1% of cited papers in oceanography in 2019]**
- o Kiss A. E. et al. (2020), ACCESS-OM2: A Global Ocean-Sea Ice Model at Three Resolutions. *Geophysical Model Development*, 13, 401-442.
- o **Holmes, R.**, Renault, L., Maillard, L. and Boucharel, J. (2024) Air-sea coupling feedbacks over Tropical Instability Waves. *Journal of Physical Oceanography*, 54, 2165-2184.
- o Huguenin, M., **Holmes, R.** and England, M. (2022) Drivers and distribution of global ocean heat uptake over the last half century. *Nature Communications*, 13, 4921.
- o Malan, N., Roughan, M., Stanley, G., **Holmes, R.** and Li, J. (2022) Quantifying cross-shelf transport in the East Australian Current System: A budget-based approach. *Journal of Physical Oceanography*, 52, 2555-2572.
- o Huguenin, M., **Holmes, R.** and England, M. (2020), Diabatic contributions to warm water volume variability over ENSO events. *J. Climate*, 33 (22), 9945-9964.
- o **Holmes R.**, McGregor, S., Santoso, A. and England, M. (2019) Contribution of Tropical Instability Waves to ENSO Irregularity, *Climate Dynamics*, 52, 1837-1855.

Competitive Grants and Funding

I have been involved in successful competitive research grant funding worth a total of ~ 2.5 million AUD, including a US National Oceanographic and Atmospheric Administration (NOAA) research grant and sole-Chief Investigator grants such as an **Australian Research Council Discovery Early Career Researcher Award** (DECRA, AUD\$440,185) and a National Computational Institute (NCI) computational grant (~AUD\$58,000).

Awards

Invited keynote talk for recent paper, DRAKKAR annual workshop	2022
<i>DRAKKAR European ocean modelling consortium</i>	<i>Grenoble, France</i>
Most selfless contribution to COSIMA award	2020
<i>Consortium for Ocean and Sea Ice Modelling in Australia (COSIMA)</i>	<i>Canberra, Australia</i>
Nominee to attend the Lindau Nobel Laureates Meeting	2017
<i>Australian Academy of Science</i>	<i>Canberra, Australia</i>
Invited participant, Physical Oceanography Dissertation Symposium	2016
<i>National Science Foundation</i>	<i>Hawaii, United States</i>

Service and Outreach (selected)

- **State of the ocean briefings**, Bureau of Meteorology monthly climate meetings, 2023-present
- **ACCESS-NRI Machine Learning for Weather and Climate Working Group Co-chair**, 2024-2026
- **Bureau Science Conversations Seminar Series organiser**, 2023-present
- **Consortium for Ocean and Sea Ice Modelling in Australia (COSIMA) leadership team member and weekly seminar convenor**, 2020-2024.
- **CMIP6 Contributor**, I performed the ACCESS-OM2-025 simulations contributed to the IPCC's CMIP6.
- **US CLIVAR working group member**, tropical Pacific decadal climate variability, 2022.
- **Expert reviewer** for research grant funding and 40+ scientific articles

Student supervision and teaching

I have co-supervised 5 Ph.D. students (4 completed, 1 in progress), 3 Masters and Honours students and several undergraduate students. These projects have led to 13 journal publications. I have contributed to teaching in 5 UNSW and University of Sydney climate and oceanography courses. My teaching has received excellent feedback from students, with anonymous survey comments such as *"Ryan knows how to teach. He explained concepts I dreaded before coming into this course in such a way that I could fully grasp them."*.

Languages and Miscellaneous

English: Native

French: Fluent - DELF B2 Diploma

Australian baseline security clearance